

Personal Summary:

I'm an experienced Senior Elixir developer with a successful track record in architecting scalable, fault-tolerant systems and managing high-performing technical teams. Having founded and exited a SaaS startup leveraging Phoenix and LiveView, I combine deep technical proficiency with entrepreneurial insight. I'm currently seeking a Senior Elixir Engineering/Technical Product Management position that enables me to leverage both my passion for hands-on coding and my experience as a startup founder, ideally in an innovative environment that values creativity and technical excellence.

Education

UNIVERSITY OF UTAH

Salt Lake City, UT

B.Sc. in Computer Science

GPA Overall: 3.7/4.0

August 2016 – May 2019

UNIVERSITY OF SOUTH ALABAMA

Mobile, AL

B.Sc. in Computer Science

GPA Overall: 3.4/4.0

August 2013 – December 2014

Technology & Skills Summary

Computer Languages: Javascript/Typescript (React, Vue), Python, Elixir

Computer Databases: PostgreSQL, Redis, Amazon DynamoDB, Mnesia/ETS

AWS/Google Cloud Systems: EC2, S3, DynamoDB, IAM, SES, SNS, ELB, Billing, Cloud SQL, Google Cloud Compute

Computer Frameworks: Ansible, Docker, Vagrant, Phoenix, Django, Git, Joomla, Ubuntu Server, Metabase, VueJS, Tailwind, TravisCI, Gitlab CI/CD

Academic Awards: Dean's List (Fall 2013, Spring 2013), Honor Award (Fall 2016, Spring 2018, Fall 2018, Spring 2019).

Elixir Libraries: Phoenix, LiveView, Oban, Ecto, Finch, Flow, ExUnit, Wallaby, Mock, Jason, Libcluster, Timex

Employment History

WorldLink Communications Ltd.

Full Stack / Principal Elixir Engineer

December 2024– Present Day

Daily Duties Included:

- Engineered an Elixir pipeline to ingest Nimble HLS Server logs to parse terabyte-scale log data on the fly, delivering sub-second realtime visualizations to 500K+ concurrent users during nightly traffic surges with <200ms tail latency.
- Architected a distributed analytics dashboard using Phoenix and LiveView to keep track of over a million devices of what is being watched on TV while parsing over 100,000/events per second. Leveraged Elixir, Flow, Genservers, Dynamic Supervisors and Libcluster on the backend for clustering to create a high-performance system.
- Designed a stateful, fault-tolerant system leveraging the OTP platform with Genservers and Dynamic Supervisors within Elixir to isolate and monitor per-user and per-device sessions, achieving 99.99% uptime under sustained 100,000/events per second loads during peak hours.
- Eliminated external dependencies (MySQL, Redis) by using Elixir's internal lookups like OTP, Mnesia, ETS, enabling a single-binary deployment that simplifies installation, reduces operational overhead, and ensures portability. Single External Dependency on PostgreSQL.

- Integrated an auto-scaling strategy for cluster of nodes (using Libcluster and Ansible) that can dynamically spin up or tear down nodes based on load, minimizing infrastructure costs while preserving high throughput.
- Partnered with network engineers, DevOps, and product teams to align analytics capabilities with business goals, significantly improving features and the UI of the dashboard.
- Developed LiveView frontend with visualizations to view active viewers, live content/channel popularity, geographic distribution and session duration.
- Implemented Flow for parallel data processing, enabling high-throughput event streaming, and transformation while minimizing CPU and memory overhead for ETL workloads.
- Leveraged Elixir's Cachex to store pre-aggregated data, avoiding repeated queries to a large PostgreSQL database and ensuring snappy frontend performance across multiple loads.
- Designed and executed load testing using Tsung to simulate 50K+ concurrent TCP connections on the Elixir/Phoenix system, validating system performance against SLAs and identifying optimization opportunities further by using Observer.
- Performed distributed tracing and metric collection with Elixir's Observer Telemetry, enabling proactive performance tuning and quick root-cause analysis across multi-node clusters.
- Developed a comprehensive testing strategy using ExUnit and integration tests to ensure system reliability, prevent defects, and validate performance under high-throughput conditions.
- Designed and automated end-to-end tests for Phoenix LiveView components using Wallaby, validating real-time UI interactions across multiple users and pages.
- Implemented LiveView hooks and client-side JavaScript interop to enhance user experience, enabling seamless intergration with external Javascript libraries for data visualization.
- Implemented pattern matching in LiveView socket event handlers and HTTP responses to dynamically process user interactions and efficiently render UI updates.
- Optimized real-time data updates in Phoenix LiveView by leveraging LiveView streams, efficiently managing memory usage on sockets and reducing DOM diff size to improve UI performance at scale.

Social Reech

Founder (Exited – Acquire)

January 2021 – July 2024

Daily Duties Included:

- Founded and successfully exited an Elixir/Phoenix-based SaaS leveraging LiveView to automate FAQ generation through behavioral analytics, facilitate real-time customer interactions via live chat, and efficiently manage customer relationships.
- Designed and architected high-performance, fault-tolerant systems using Phoenix/Elixir, PostgreSQL, and LiveView, ensuring 99.99% uptime for over 100,000 users. Achieved a 30% reduction in server costs by transitioning from Python/Django to Phoenix/Elixir.
- Implemented intuitive Excel-based CRM and issue-tracking solutions, significantly improving communication efficiency with bank employees and eliminating training overhead through user-centric design.
- Recruited, mentored, and led a high-performing technical team, fostering a collaborative culture emphasizing innovation, accountability, and continuous improvement, directly contributing to increased productivity and revenue growth.
- Streamlined software release cycles from monthly to weekly by developing robust CI/CD pipelines using GitHub Actions, Docker, ExUnit, and Cypress automated testing frameworks, resulting in a 40% decrease in production errors.
- Delivered 12 major product modules within 18 months by effectively aligning customer requirements with internal design, engineering, and QA processes teams. Achieved a 40% reduction in feature development timelines through iterative prototyping and dedicated staging environments using Ansible.
- Achieved a 90% retention rate among enterprise clients through proactive engagement and customer-centric product discovery initiatives with 50+ financial institutions, translating complex requirements into high-impact features like RBAC, Audit Trails, multilingual support, and enhanced live chat capabilities.

- Led regular client feedback sessions, utilizing in-depth interviews and direct user insights to inform strategic product decisions, prioritize features effectively, and significantly enhance user satisfaction and feature adoption.
- Developed a data-driven product roadmap by leveraging Metabase analytics, delivering detailed reporting on user engagement, revenue patterns, and feature interactions, empowering clients with actionable insights for improved decision-making.
- Collaborated cross-functionally with Developers, UX/UI Designers, and QA Engineers, translating intricate business requirements into precise technical specifications, ensuring seamless, timely, and high-quality feature delivery.
- Used Oban to manage complex background jobs and workflows, significantly enhancing system reliability and operational efficiency. For example, utilized Oban to automate daily aggregation of customer analytics data, reducing processing time and ensuring timely delivery of actionable insights.
- Leveraged Oban to automate daily reporting workflows, including daily analytics reporting and notifications to enterprise customers, achieving a 50% reduction in manual operational tasks and enhancing overall team productivity.
- Built cross-compiled Elixir binaries using Docker for multiple environments like x86 and ARM to create dependency-free Windows/Linux executables for bank deployments, addressing strict security compliance requirements while reducing client onboarding setup time by 80% through simplified installation processes.
- Designed and optimized multi-tenant database architecture using Ecto and PostgreSQL, enabling seamless data isolation and scalability for enterprise clients across 50+ financial institutions.
- Built Elixir releases via cross-compilation to ARM/x86 binaries using Docker, enabling seamless deployment across bank environments with heterogeneous infrastructure.
- Designed a modular and reusable stateful and stateless LiveView component architecture, simplifying UI development and ensuring consistency across the dashboard.
- Designed and implemented Sentry error tracking and performance monitoring across Phoenix/LiveView applications, ensuring rapid detection and resolution of production issues.

CellPay

Full-Stack Product Manager

January 2020– December 2021

Daily Duties Included:

- Pioneered a manual payment processing system ahead of delayed Third-Party Utility API integration, enabling customers to seamlessly submit water utility payment details through a Flutter app for real-time, agent-assisted transactions at billing counters. Increased customer base by 2x within 6 months by providing an exclusive market solution while reducing third-party API dependency, and accelerated time-to-market by 70% compared to competitors.
- Engineered a dynamic Flutter application facilitating real-time service additions and modifications via a hosted JSON configuration file, eliminating the necessity for codebase alterations or app store updates. Accelerated feature launch timelines to within 1 day (compared to the industry average of over 2 weeks after receiving API from third party provider) by effectively collaborating with backend teams to streamline API structures and data workflows, positioning the solution as first-to-market and boosting customer acquisition rates by 40%.
- Developed Metabase dashboard enabling the business team to efficiently analyze daily revenue metrics, identify top-performing merchant transactions, and track customer loyalty patterns.
- Redesigned and optimized database schemas for PostgreSQL (SQL) and Redis (NoSQL), significantly enhancing system performance and reducing report loading times for internal tools by introducing caching and data aggregation techniques.
- Revamped previously slow-performing financial reporting systems by integrating Redis for caching frequently accessed data, resulting in improved scalability, faster report generation, and real-time financial insights.
- Actively participated in code reviews, consistently providing constructive feedback, ensuring adherence to best coding practices, and maintaining high-quality code standards across the development team.